i.e.

$$g(\boldsymbol{q}, t) = 0. \tag{4.3}$$

Note that by differentiating (4.3) with respect to time, we obtain $\nabla_{\boldsymbol{q}} g(\boldsymbol{q}, t) \cdot \dot{\boldsymbol{q}} + \partial g / \partial t = 0$ which involves velocities. The definition of holonomic constraints thus includes equations which relate positions and momenta, but only where these are, through the differential equations, equivalent to equations involving positions only (4.3). For more details on constrained dynamics, see for example [67, 79, 83, 164, 227, 249].

An important type of holonomic constraint arising in molecular dynamics relates to bond vibration. Recall that the stretch of a chemical bond between two atoms gives rise to a potential energy contribution of the form

$$\varphi(r) = \frac{k}{2}(r - r_0)^2,$$

where r would typically represent the distance between a pair of atoms, say atoms i and j, $r = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$. The strong chemical bond stretches involving hydrogens and various heavier atoms such as nitrogen and carbon give rise to the most rapid modes present in the atomic motion. As we have seen, a harmonic oscillator with frequency Ω, when solved using the Verlet method, has a stability restriction of the form

$$h < \frac{2}{\Omega}.$$

One approach to addressing these stiff terms is to use a multiple timestepping method. A more direct approach is to introduce constraints to simply remove the stiff bond stretches. This makes sense if (a) the motion of the constrained system can be simulated using larger timesteps and (b) the rigidification of certain vibrational terms does not significantly alter the thermodynamic or dynamic properties of interest to the modeller.

To illustrate the progression from vibrational model to constrained model, one may consider a single particle subject to a potential $U(\boldsymbol{q})$ as well as an additional stiff restraint. We suppose the restraint to have an associated coefficient $k = \varepsilon^{-1}$, where ε is a small parameter. Then we have a generalized flexible pendulum with Hamiltonian $H = \|\boldsymbol{p}\|^2/(2m) + U(\boldsymbol{q}) + \varepsilon^{-1}(\|q\| - r_0)^2/2$ (see the left panel of Fig. 4.4). The differential equations describing this model are

$$\begin{aligned} \dot{\boldsymbol{q}} &= m^{-1}\boldsymbol{p}, \\ \dot{\boldsymbol{p}} &= -\nabla U(\boldsymbol{q}) - \nabla \varphi_\varepsilon(\boldsymbol{q}), \end{aligned}$$

where the restraint potential $\varphi_\varepsilon(\boldsymbol{q}) = \varepsilon^{-1}(\|q\| - r_0)^2/2$ has associated restoring force

$$\boldsymbol{F}_\varepsilon = -\nabla \varphi_\varepsilon(\boldsymbol{q}) = \varphi_\varepsilon(\boldsymbol{q}).$$